Early Black-Hole Growth Constraints from JWST

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Abstract

Aims. Massive black holes in the early Universe must grow within limited cosmic time. We identify which JWST black-hole candidates most strongly constrain early growth models and which assumptions control these constraints. **Methods.** We compile 350 literature measurement records from 32 source families at $z \geq 4$. We analyse 224 objects with broad hydrogen-line mass estimates and include 10 additional objects in a broader comparison. We propagate reported mass uncertainties and test mass-scale shifts and alternative growth assumptions. **Results.** For $100\,M_{\odot}$ seeds formed at $z = 30$, radiative efficiency $\epsilon = 0.1$, and no merger boost, 12 of the 224 objects require a lifetime-average Eddington ratio $\overline{f_{\mathrm{Edd}}}_{\mathrm{req}} > 1$. Of these, 6 exceed this threshold in at least 95% of trials sampling their quoted mass errors under fixed growth assumptions. For UNCOVER-20466, COSMOS3D-13852 and RUBIES-EGS-55604, at least 95% of trials still require an average Eddington ratio above one after reducing each mass by 0.5 dex (a factor of approximately 3.2). Uncertainties in the mass estimators also affect these constraints. Delaying seeding to $z = 20$ makes GN-z11 the most demanding object. For A2744-QSO1, an independent dynamical mass estimate raises the required average Eddington ratio to approximately one. **Conclusions.** At $\epsilon = 0.05719$, the ideal thin-disc efficiency for a non-spinning black hole, all 234 central mass estimates are reachable from $100\,M_{\odot}$ seeds formed at $z = 30$ without lifetime-average accretion above the Eddington limit.

Key words. black hole physics – accretion, accretion disks – galaxies: active – galaxies: high-redshift – quasars: supermassive black holes

1 Introduction

Massive black holes observed in the early Universe must have assembled within the available cosmic time. The strength of this constraint depends on the observed black-hole mass and redshift and on assumptions about seed formation, radiative efficiency, and the duration of active accretion. Different seed and efficiency assumptions can therefore give different required accretion rates for the same object.

We use a literature-based catalogue of JWST-identified high-redshift accreting black holes and candidates to test which combinations of seed mass, formation time, radiative efficiency and accretion rate can reproduce the observed masses by the observed redshifts. The atlas covers $z \geq 4$, extending from the first billion years to a cosmic age of approximately 1.54 Gyr at its lower-redshift boundary. We calculate the average accretion rates required across a range of seed and growth assumptions, incorporating reported mass uncertainties and accounting for differences in detection evidence and mass-estimation methods. We retain published measurements, object identity links and mass-estimation sources so that measurements from different studies can be compared.

Analytic seed-growth comparisons already establish the trade-off between seed mass and sustained accretion. For example, P. Dayal (2024) considered light and heavy seeds at $z_{\mathrm{seed}} = 25$ and explored a primordial-seeding interpretation. We apply the same growth equations to a larger catalogue and test how each object's constraints depend on its published mass, measurement method and growth assumptions. We calculate whether each black hole could reach its observed mass while accreting, on average, at or below the Eddington limit. We repeat this calculation using masses drawn from the published uncertainty intervals, uniform shifts in the mass estimates, and alternative published measurements. These tests show how the inferred accretion rate depends on mass uncertainty and the method used to estimate the mass. For $100\,M_\odot$ seeds formed at $z = 30$, radiative efficiency $\epsilon = 0.1$ and no mass added by mergers, 12 objects in our Balmer-line sample require an average Eddington ratio above one. We list all 12, describe the limitations reported for their mass estimates, and propose observations to test those estimates. For four objects also studied by Dayal, we calculate separately how the required accretion rates change when we update the measured masses and redshifts and when we start growth earlier. For A2744-QSO1, we repeat the calculation with an independent dynamical mass estimate. We keep the original mass estimate in the catalogue and report the dynamical comparison separately. The JADES broad-line census (I. Juodžbalis et al. 2026a) measures current accretion and host scaling relations, and the narrow-line census (J. Scholtz et al. 2025) identifies candidates for obscured activity. We use these measurements to calculate the cumulative growth needed to reach the adopted masses. The contributing studies targeted different kinds of objects and reached different detection limits. The probability that a black hole would be selected and detected therefore varies between studies and is unknown for the combined sample. The source-family inventory is given in Appendix A.

1.1 Theoretical framework

We use the luminosity-based Eddington ratio $f_{\rm Edd} = L_{\rm bol}/L_{\rm Edd}$ and a fixed radiative efficiency ϵ, the fraction of accreted rest-mass energy emitted as radiation. For the available growth interval,

$$\Delta t = t(z_{\rm obs}) - t(z_{\rm seed}) > 0, \qquad \overline{f_{\rm Edd}} = \frac{1}{\Delta t}\int_{t(z_{\rm seed})}^{t(z_{\rm obs})} f_{\rm Edd}(t)\,dt. \tag{1}$$

The analytic growth relation adopted by P. Dayal (2024) gives

$$M_{\rm BH}(t_{\rm obs}) = B_{\rm merge} M_{\rm seed} \exp\left[\frac{1-\epsilon}{\epsilon}\frac{\overline{f_{\rm Edd}}\,\Delta t}{t_{\rm Edd}}\right], \tag{2}$$

where $t_{\rm Edd} \simeq 0.45$ Gyr. The merger multiplier is $B_{\rm merge} = 1$ for the reference calculation and 2 for a factor-of-two mass boost. We represent mergers by multiplying the seed mass by a constant factor before calculating accretion. This approximation omits the timing and sequence of mergers, their mass ratios, and mass lost through gravitational radiation. The required lifetime-average ratio is

$$\overline{f_{\rm Edd}}_{\rm req} = \max\left[0, \frac{\epsilon}{1-\epsilon}\frac{t_{\rm Edd}}{\Delta t}\ln\left(\frac{M_{\rm BH}}{B_{\rm merge}M_{\rm seed}}\right)\right]. \tag{3}$$

If $B_{\rm merge}M_{\rm seed}$ exceeds the observed black-hole mass, the logarithm is negative and we set the required average Eddington ratio to zero. Even with zero accretion, this seed and merger contribution predict a mass larger than observed. Reproducing the observed mass would require a smaller seed, a smaller merger contribution, or a model that includes mass loss. This requirement averages accretion over the full growth interval. Inferring a duty cycle also requires assumptions about the rates during active and quiet phases.

We calculate cosmic ages using $H_0 = 67.3\,{\rm km\,s^{-1}\,Mpc^{-1}}$ and $\Omega_m = 0.315$, matching these two values in P. Dayal (2024), who cite Planck. We impose spatial flatness and set $\Omega_\Lambda = 1 - \Omega_m = 0.685$. The age calculation includes matter and a cosmological constant and neglects radiation. Appendix B gives the supporting relations, implementation and inverse seed-mass diagnostic. Equations 2–3 assume constant radiative efficiency. The ideal thin-disc efficiencies used in Figure 1 are 0.03775, 0.05719 and 0.42265 for dimensionless spins $a = -1, 0, +1$, respectively (J. M. Bardeen et al. 1972). Each track holds efficiency constant; spin evolution is unspecified.

2 Catalogue construction

We compile 350 literature measurements from 32 source families at $z \geq 4$. Each adopted mass is linked to its source, method and reported uncertainty. We select one published measurement per object using the source assessments and retain the alternatives for sensitivity tests.

Growth analysis requires a numerical mass and redshift, an identified mass method and a resolved lensing treatment. Upper limits, indirect model ranges and masses inferred by assuming an Eddington ratio are excluded from numerical inference. We also exclude all six records associated with three unresolved identity groups, including three records with eligible masses. This leaves 224 primary objects and an inclusive exploratory sample of 234 objects.

For the main comparison, we use masses estimated from broad Hα or Hβ emission lines in objects with secure or probable evidence for an accreting black hole, using the evidence classifications recorded from the source papers. Seven masses are explicitly flagged as conditional on a broad-line-region interpretation: six candidates with intermediate-width Hα emission and possible outflow contamination (W. Ren et al. 2025), and RUBIES-EGS-49140 (A. J. Taylor et al. 2025), for which an alternative non-AGN interpretation is recorded. These seven objects enter the expanded sample only. After the identity exclusions, the main comparison contains 224 objects, labelled primary in the tables and figures. Other limitations of a virial mass, including possible scattering, are recorded separately. COSMOS3D-13852 remains in the main sample on the basis of its published broad-line AGN identification; we test the reported scattering bias in Section 5.2. These single-epoch virial estimators combine the line width, which estimates gas velocity, with a luminosity-based estimate of the broad-line region's radius. Their calibration is based on reverberation mapping of nearby AGN (Y. Shen 2013). Using the same line family and mass-estimation framework reduces differences between methods in the comparison. Applying these calibrations at high redshift still requires assessing gas dynamics, geometry, obscuration and lensing for each object. We also analyse an expanded sample of 234 objects, labelled exploratory in the tables and figures. It includes the 224 primary objects, nine additional candidates, and GN-z11. GN-z11's mass is inferred from rest-frame ultraviolet emission lines using local virial relations (R. Maiolino et al. 2024). We include it in the expanded comparison because its lines and calibration differ from those of the Balmer-line sample. The sample division reflects the evidence for accretion and the methods being compared; the reliability of each mass estimate requires an individual assessment. When a paper gives a mass without an uncertainty, we can calculate the average Eddington ratio needed to reach that mass. We cannot calculate the probability that this ratio exceeds one from the published information, because no mass uncertainty distribution is available. We therefore include these objects in calculations at their published masses and omit them from probability calculations. The mass estimates retained in each sample still use different calibrations, and the contributing studies have different target-selection criteria and detection limits. Appendix A gives the source inventory, identity decisions and sample accounting.

3 Growth-model comparison

3.1 Reference model

We calculate the required lifetime-average Eddington ratio using Eq. 3, with $M_{\rm seed} = 10^2 M_\odot$, $z_{\rm seed} = 30$, $\epsilon = 0.1$ and $B_{\rm merge} = 1$ as the reference history. We rank objects by this required average Eddington ratio. Figure 1 illustrates tracks at four fixed efficiencies; tracks are selected for proximity to the observed masses. Appendix B.5 specifies the display selection.

3.2 Uncertainty and sensitivity

We use Monte Carlo sampling to propagate reported asymmetric log-mass errors, with 10,000 draws per object. Each draw has equal probability of falling above or below the central mass, with its distance drawn from a half-normal distribution scaled by the corresponding upper or lower error. Redshifts are held fixed. For each sampled mass, we calculate the average Eddington ratio needed to grow from the assumed seed to that mass. The fraction of the 10,000 Monte Carlo draws with a required ratio above one estimates the probability that growth requires an average rate above the Eddington limit, given the adopted mass-error distribution and growth assumptions. Probability calculations include 212 primary and 222 expanded-sample objects; the 12 objects without reported errors enter only calculations at their published masses.

We test mass-scale sensitivity by shifting all adopted log masses and their error draws by offsets of -0.5, -0.3, 0, $+0.3$ and $+0.5$ dex, keeping the sample and growth assumptions fixed. These fixed offsets are stress tests at the uncertainty scale of single-epoch mass estimates (Y. Shen 2013). For each offset, we report the fraction of shifted Monte Carlo draws that require an average Eddington ratio above one. Each probability assumes that particular mass shift and the stated seed mass, starting redshift, radiative efficiency and merger contribution. Sampling and conversion details are given in Appendix B.5.

4 Results

4.1 Required accretion under the reference assumptions

For a $10^2\ M_\odot$ seed under the reference history, the 224-object primary sample has median required $f_{\mathrm{Edd}} = 0.578$ (interquartile range 0.487–0.682), with 12 point requirements above unity and none above two. The inclusive 234-object sample has median 0.574 (0.488–0.679) and 14 above unity. UNCOVER-20466 is most demanding: its point requirement is 1.452 and its reported-error median is 1.453 (16th–84th percentiles 1.354–1.549). Table 1 lists the 12 primary threshold objects; their ordering by point requirement differs from ordering by quoted-error probability.

The shorter growth time at $z = 10.603$ places GN-z11 second in the required-rate ranking despite its smaller mass. Its UV virial estimator places it in the exploratory sample.

4.2 Robustness to reported mass uncertainty

Of the 12 primary point requirements above unity, eight remain above it at the 16th percentile and six have conditional $P(\overline{f_{\mathrm{Edd}}}_{\mathrm{req}} > 1) \geq 0.95$ (Fig. 2). The exploratory counts are 14 at the median, ten at the 16th percentile and eight at $P \geq 0.95$; 15 exceed unity by the 84th percentile. Probabilities follow Section 3.2 and use 212 primary and 222 exploratory objects with reported errors. The 12 primary objects without errors enter only point counts.

4.3 Sensitivity to the virial-mass scale

Downward mass shifts of 0.3 and 0.5 dex reduce the primary point/16th-percentile/ $P \geq 0.95$ counts from 12/8/6 to 9/6/4 and 6/4/3, respectively (Table 2). UNCOVER-20466, COSMOS3D-13852 and RUBIES-EGS-55604 retain $P \geq 0.95$ at -0.5 dex. Biases specific to each source may exceed this offset (Section 5.2), and the required rates still depend on seed mass, formation time and radiative efficiency.

Replacing the redshifts, masses and errors of 44 Baccus counterparts matched by identifier and confirmed by position with published Table 1 values (C. J. Baccus & X. Xu 2026) preserves the primary 12/8/6

Table 1: The 12 primary objects above the reference threshold, ordered by their point requirements. Masses and errors are in $\log_{10}(M/M_{\odot})$; all estimators are single-epoch virial. f_0 is the reference point requirement, f_{16} its reported-error 16th percentile, and $P_0 = P(\overline{f_{\mathrm{Edd}_{\mathrm{req}}}} > 1)$. f_- and P_- apply the fixed -0.5 dex offset. Δm (dex) is the reduction in log mass that brings the central required average Eddington ratio to one, with the growth assumptions unchanged. All probabilities use the quoted mass errors and fixed growth assumptions. Additional limitations and proposed observations are in Table 6.

Object/source	Line	Log mass	f_0	f_{16}	P_0	f_-	P_-	Δm
UNCOVER-20466 J. E. Greene et al. (2024)	Hβ	$8.17^{+0.42}_{-0.42}$	1.452	1.354	> 0.999	1.335	> 0.999	1.921
GS-20057765 I. Juodžbalis et al. (2026a)	Hβ	$7.33^{+0.62}_{-0.70}$	1.355	1.177	0.980	1.228	0.901	1.397
COSMOS3D-13852 X. Lin et al. (2026)	Hβ	$8.60^{+0.35}_{-0.33}$	1.314	1.247	> 0.999	1.214	> 0.999	1.576
RUBIES-EGS-55604[a] D. D. Kocevski et al. (2025)	Hβ	$9.31^{+0.10}_{-0.10}$	1.267	1.250	> 0.999	1.180	> 0.999	1.541
CEERS-1019 R. L. Larson et al. (2023)	Hβ	$6.95^{+0.37}_{-0.37}$	1.205	1.115	0.988	1.083	0.822	0.842
GS-20030333 I. Juodžbalis et al. (2026a)	Hβ	$7.42^{+0.65}_{-0.48}$	1.133	1.033	0.907	1.029	0.613	0.637
GS-164055 I. Juodžbalis et al. (2026a)	Hβ	$7.63^{+0.59}_{-0.66}$	1.065	0.941	0.696	0.970	0.395	0.342
GN-38509[b] I. Juodžbalis et al. (2026a)	Hα	$8.57^{+0.37}_{-0.38}$	1.065	1.005	0.860	0.984	0.392	0.398
J0910_2028_12910[a] C. J. Baccus & X. Xu (2026)	Hα	$8.58^{+0.01}_{-0.01}$	1.053	1.051	> 0.999	0.973	< 0.001	0.330
CEERS-00717 Y. Harikane et al. (2023)	Hα	$7.99^{+0.16}_{-0.17}$	1.027	0.998	0.821	0.942	0.018	0.160
ZS7[c] H. Übler et al. (2024)	Hβ	$7.70^{+0.40}_{-0.40}$	1.023	0.954	0.625	0.934	0.173	0.130
GN-4685 I. Juodžbalis et al. (2026a)	Hβ	$7.36^{+0.45}_{-0.42}$	1.017	0.938	0.585	0.922	0.181	0.091

[a] A separate 0.5-dex calibration scale is reported for RUBIES-EGS-55604 and J0910_2028_12910; [b] a separate 0.3-dex scale is reported for GN-38509. These scales are omitted from the quoted errors and probabilities shown here. [c] ZS7's quoted 0.4-dex error already includes calibration scatter. Unmarked rows have no separate numerical calibration scale recorded; their quoted errors may omit other mass-estimator uncertainties.

Table 2: Fixed mass-scale stress tests under the reference growth history. Triplets give counts above unity at the point estimate, at the 16th percentile, and with conditional $P(\overline{f_{\mathrm{Edd}_{\mathrm{req}}}} > 1) \geq 0.95$. The probability-based counts use 212 primary and 222 exploratory objects with reported errors; point counts use all 224 and 234 objects.

Mass offset (dex)	Primary	Exploratory
-0.5	6 / 4 / 3	7 / 5 / 4
-0.3	9 / 6 / 4	10 / 7 / 5
0	12 / 8 / 6	14 / 10 / 8
$+0.3$	15 / 13 / 12	17 / 15 / 14
$+0.5$	19 / 16 / 13	21 / 18 / 15

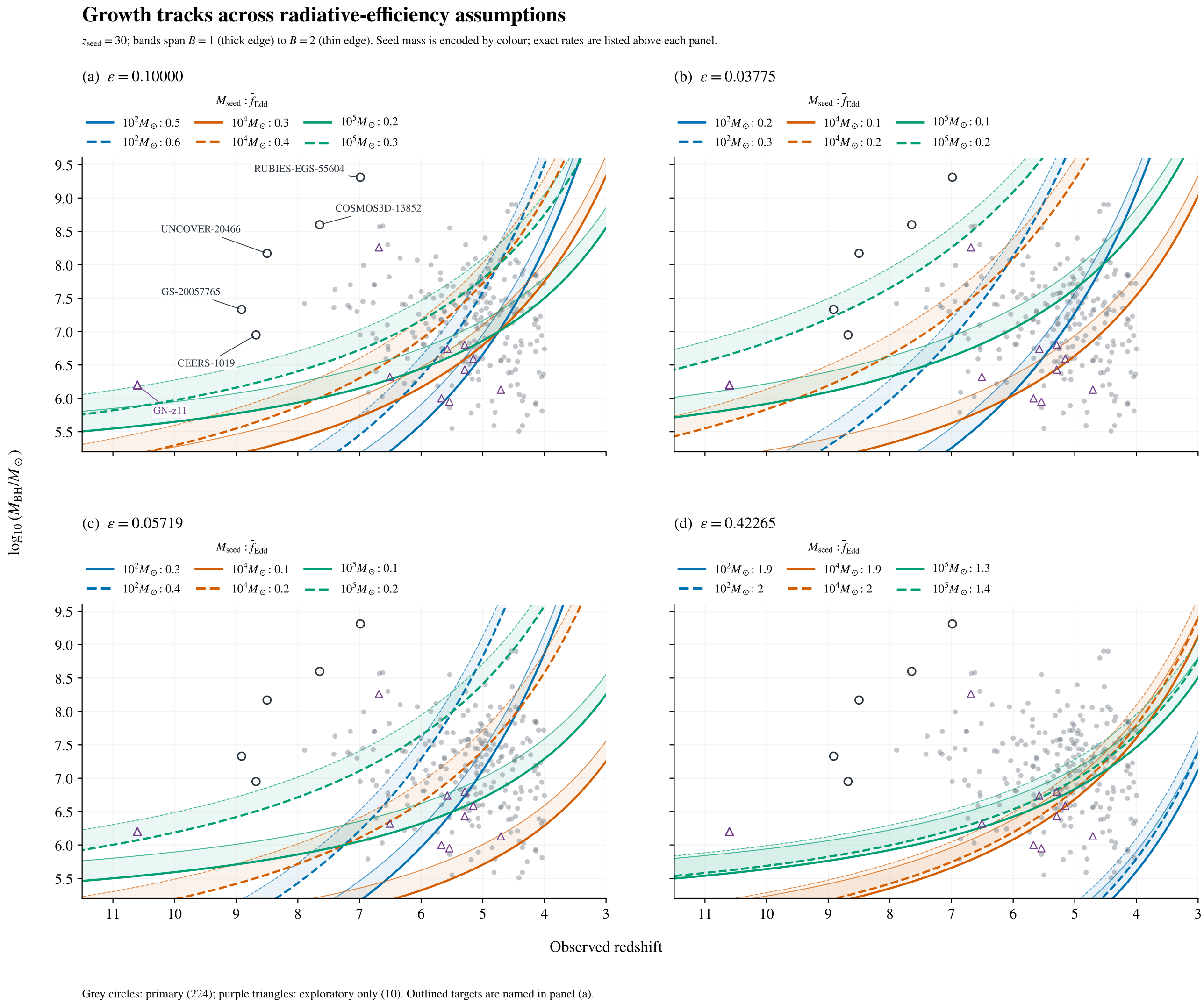

Figure 1: Growth tracks with 224 primary and ten exploratory-only objects. Panels show four constant radiative efficiencies at $z_{\rm seed} = 30$; colours distinguish seeds of 10^2, 10^4 and $10^5 M_\odot$, and legends list the selected average Eddington ratios. Shaded bands span merger boosts $B = 1$–2, with thick and thin edges respectively. The bands show the effect of the assumed merger boost. Six named objects are outlined in every panel. The reversed redshift axis spans $z = 11.5$ to $z = 3$. Excluded identities and objects without eligible masses are omitted.

and exploratory 14/10/8 counts and both top-five rankings (Table 3). Of five unmatched records, one is identity-excluded; retaining or omitting the other four leaves these results unchanged (220/230 objects when omitted). The test updates the matched measurements while keeping all six identity exclusions, method classes, growth parameters and sampling rules fixed.

4.4 Model compatibility and measurement choice

Increasing the seed mass to $10^3\,M_\odot$ reduces the primary count above unity from 12 to four; $10^5\,M_\odot$ seeds remove all point exceedances. Delaying seeding to $z = 20$ raises the primary count to 22, while $\epsilon = 0.05719$ removes all point exceedances in both samples (Table 4). Each test changes one assumption: seed mass (P. Dayal 2024), available growth time, or the efficiency of an ideal thin disc around a non-spinning black hole (J. M. Bardeen et al. 1972).

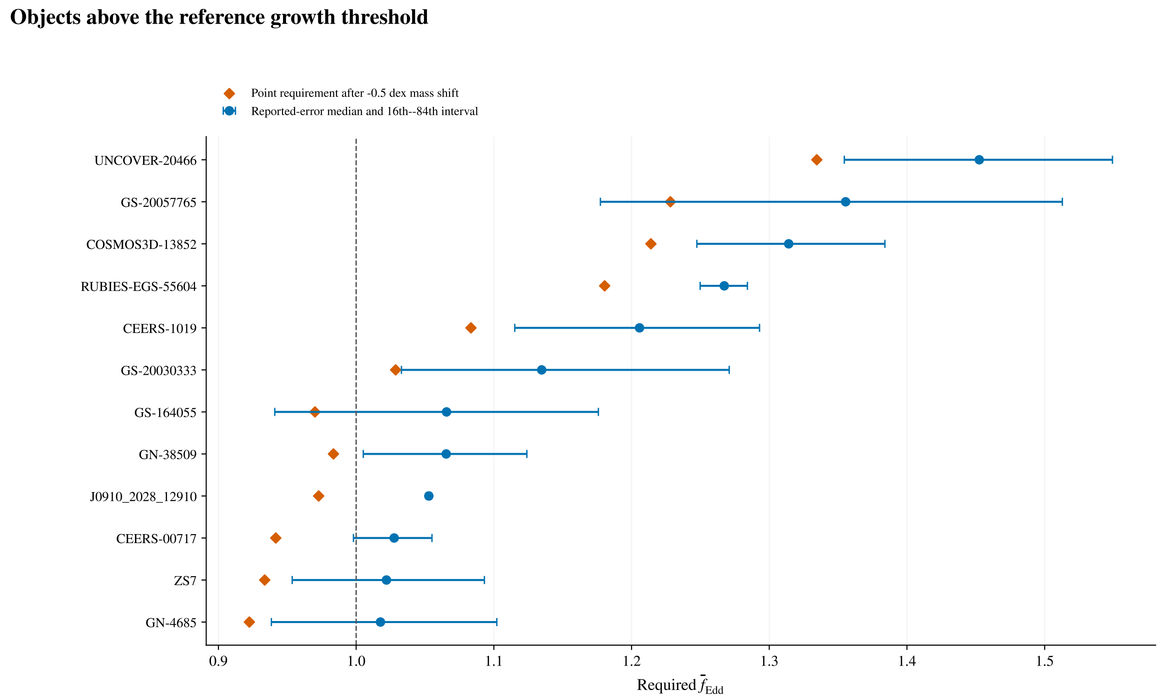

Figure 2: The 12 primary objects whose reference point requirements exceed unity. Circles and bars show reported-error medians and 16th–84th percentile intervals; diamonds show point requirements after a −0.5 dex mass shift. The vertical line is the reference threshold. The selected objects all have reported errors; these intervals do not include additional estimator systematics.

Table 3: Effect of adopting the published Baccus measurements after excluding unresolved identities. Every row excludes all open identity groups. Counts denote point requirements above unity, 16th percentiles above unity, and conditional $P \geq 0.95$, respectively. The adopted catalogue values define the baseline.

Sample	Measurement treatment	Objects	Counts
Primary	Frozen values	224	12/8/6
Primary	Published; retain unmatched	224	12/8/6
Primary	Published; omit unmatched	220	12/8/6
Exploratory	Frozen values	234	14/10/8
Exploratory	Published; retain unmatched	234	14/10/8
Exploratory	Published; omit unmatched	230	14/10/8

Removing the six ambiguous records from the original 227/237-object sets preserves threshold counts and top-five rankings in every listed scenario, including the reference probability counts. The primary median changes from 0.574 to 0.578; the objects above the reference threshold remain the same.

Figure 3 shows the maximum fixed efficiency allowing $\overline{f_{\rm Edd}}_{\rm req} \leq 1$ for five targets as a function of seed mass. The $z_{\rm seed} = 20$ and 30 curves show GN-z11's greater sensitivity to the starting time. These boundaries use central mass estimates and assume no merger boost.

Appendix C gives the fractions of objects whose masses can be reproduced from each seed-mass range under specified accretion rates, spin-dependent efficiencies and merger contributions.

Seven alternate-measurement comparisons across six objects (two for CEERS-2782) span mass shifts of −0.600 to +0.949 dex and required-$f_{\rm Edd}$ shifts of −0.083 to +0.121 (Fig. 4). Both estimates remain below unity in every pair. This comparison covers six objects below the threshold. Testing the objects above it

Table 4: Point-estimate requirements above $\overline{f_{\rm Edd}} = 1$ under fixed histories. Each row changes only the named parameter from the reference ($M_{\rm seed} = 10^2 M_\odot$, $z_{\rm seed} = 30$, $\epsilon = 0.1$, $B_{\rm merge} = 1$). Counts use central mass estimates and provisional object identities.

Scenario	Primary (224)	Exploratory (234)
Reference	12	14
$M_{\rm seed} = 10^3 M_\odot$	4	5
$M_{\rm seed} = 10^5 M_\odot$	0	0
$z_{\rm seed} = 20$	22	24
$\epsilon = 1 - \sqrt{8/9}$	0	0

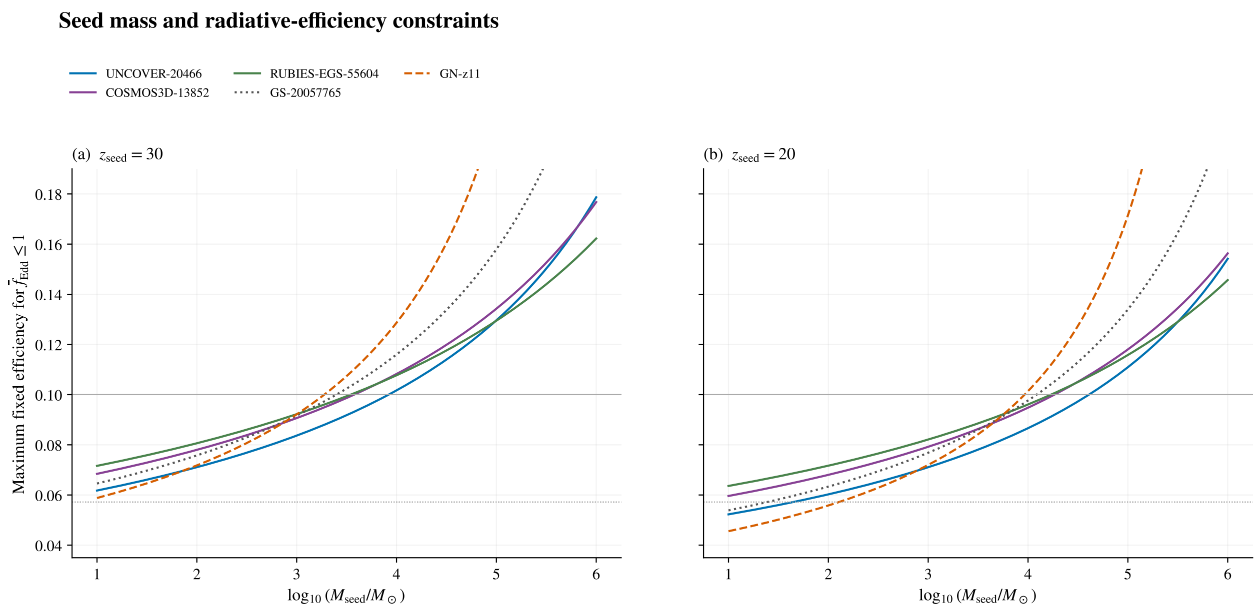

Figure 3: Seed-mass–efficiency boundaries for Eddington-limited average growth at two starting redshifts. Each curve uses an adopted point mass and no merger boost. Below a curve the required average is at most unity. Horizontal guides mark $\epsilon = 0.1$ (solid grey) and $1 - \sqrt{8/9}$ (dotted grey). GN-z11 (dashed) is exploratory; the other objects are primary. Each curve holds efficiency constant.

requires additional independent mass estimates.

4.5 Separating measurement and starting-time changes

For four unambiguous matches to Table 1 of P. Dayal (2024), Table 5 first replaces the earlier masses and redshifts with our adopted values at $z_{\rm seed} = 25$, then changes seeding to $z = 30$. All columns use our cosmology, $\epsilon = 0.1$, $10^2 M_\odot$ seeds and no merger boost. Earlier seeding lowers the requirements more than the measurement updates change them. The comparison isolates these two effects for the four matched objects using our growth calculation.

4.6 External dynamical-mass check: A2744-QSO1

For A2744-QSO1, we compare the virial mass adopted in the catalogue $\log_{10}(M/M_\odot) = 7.3 \pm 0.2$ (J. E. Greene et al. 2024) with the independent MOKA3D dynamical estimate 7.7 ± 0.3 (I. Juodžbalis et al.

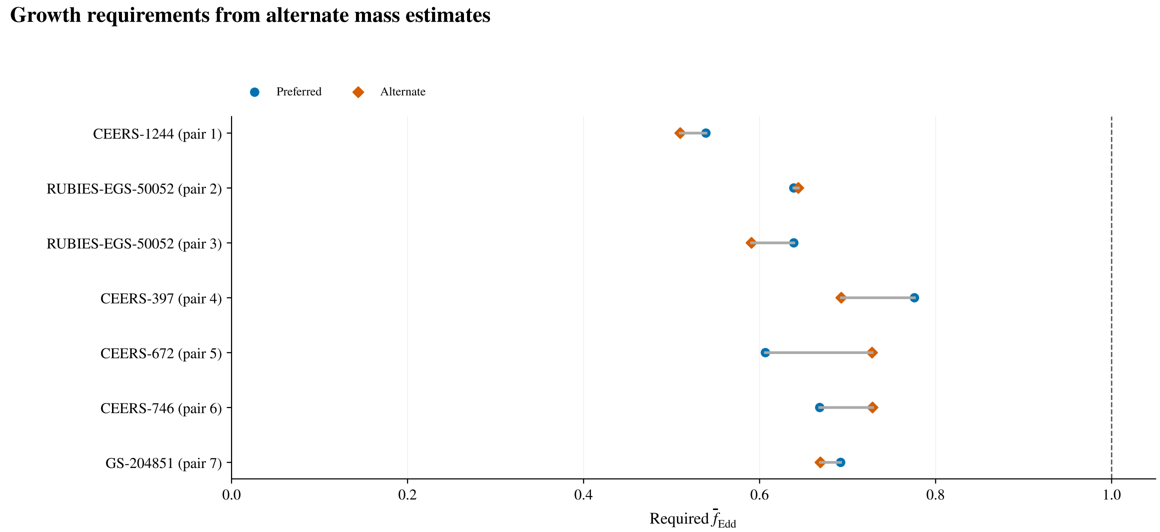

Figure 4: Preferred (circles) and alternate (diamonds) point requirements for seven measurement pairs across six primary objects. Pair numbers follow the released alternate-measurement table, which supplies stable measurement IDs; RUBIES-EGS-50052 (CEERS-2782) has two alternatives. All pairs remain below the reference threshold.

Table 5: Measurement and starting-time comparison for four objects shared with the Dayal study. Columns give required average $f_{\rm Edd}$; the first two share $z_{\rm seed} = 25$. The analysis files provide the input measurements and their source locations.

Object	Earlier inputs, 25	Adopted inputs, 25	Adopted inputs, 30
GN-z11	1.577	1.578	1.437
A2744-QSO1	0.974	0.974	0.929
UNCOVER-20466	1.540	1.548	1.452
CEERS-00717	1.074	1.076	1.027

2026b), using $z = 7.04$ and the reference history. We retain the values from arXiv:2508.21748v2; the reference lists the subsequent journal publication. The dynamical estimate depends on lens reconstruction and inclination. We propagate the quoted two-sided mass error; the separate inclination-free lower limit is a one-sided constraint. Using the central mass estimates, the required average Eddington ratio rises from 0.929 to 1.000. With a symmetric normal approximation to the quoted log-mass errors, the exact 16th–84th percentile intervals are 0.895–0.964 and 0.947–1.052, respectively. For the dynamical mass, the unrounded required ratio is just below one. Its uncertainty interval includes average accretion rates both below and above the Eddington limit. With this mass-error distribution and the growth assumptions held fixed, the probability that the required average Eddington ratio exceeds one is 0.497. This comparison tests the mass of one catalogue object. Calibrating the virial sample would require independent measurements for more objects.

5 Discussion

5.1 Dependence on seed and growth assumptions

The objects above the reference threshold lie at $z = 6.6214$–10.603; most catalogue objects lie at lower redshift and serve as the comparison sample. Their growth requirements constrain combinations of seed mass, starting time, efficiency and accretion. Several formation histories satisfy these constraints.

At fixed seed, growth interval and merger factor, required $f_{\rm Edd}$ scales as $\epsilon/(1-\epsilon)$. Lowering ϵ from 0.1 to 0.05719 multiplies all requirements by 0.54594: the largest becomes 0.793. The requirement for super-Eddington growth therefore depends on the assumed efficiency. Several spin and seed combinations can reproduce the same mass. Larger seed masses, earlier growth onset and merger contributions also reduce the required accretion rate.

Table 6 lists observations that could test the mass estimates of the objects above the reference threshold. Independent masses and constraints on scattering, outflows, obscuration and lensing can distinguish these interpretations. We propose these observations based on the known limitations of each mass estimate.

5.2 Estimator credibility and observational tests

Some mass-estimator biases may exceed the −0.5 dex shift tested above. UNCOVER-20466's demagnified Hβ mass depends on lensing and broad-line interpretation (J. E. Greene et al. 2024); RUBIES-EGS-55604's dust-corrected Hβ mass has a separately recorded 0.5 dex calibration scale (D. D. Kocevski et al. 2025). For COSMOS3D-13852, scattering could inflate the Hβ mass by 1–2 dex (X. Lin et al. 2026). Applying downward offsets of 1 and 2 dex gives point requirements of 1.115 and 0.916, respectively. These stress tests use the bias range discussed by the source; the actual correction remains uncertain.

The mass reductions in Table 1 show how far each central mass lies above the value reachable at an average Eddington ratio of one. We subtract the log mass predicted by Eq. 2 at $\overline{f_{\rm Edd}} = 1$ from the observed log mass, keeping the seed, starting redshift, efficiency and merger contribution fixed. For J0910_2028_12910, a reduction of 0.330 dex is sufficient. This is smaller than its separately reported 0.5-dex calibration scale (C. J. Baccus & X. Xu 2026), although its 0.01-dex quoted error gives $P > 0.999$ in the reported-error calculation. COSMOS3D-13852 requires a reduction of 1.576 dex, within the 1–2 dex range of possible scattering bias. UNCOVER-20466 requires 1.921 dex; its lensing and broad-line assumptions still require independent tests. These mass reductions follow from the adopted growth parameters. ZS7's quoted 0.4 dex already includes calibration scatter (H. Übler et al. 2024). The probabilities therefore reflect different combinations of measurement and calibration uncertainty across the sample.

5.3 Growth constraints for individual objects

UNCOVER-20466 and GS-20057765 have reference requirements of 1.452 and 1.355, with 16th percentiles above unity. Their point requirements remain above unity for $10^3 M_\odot$ seeds (1.217 and 1.101), but fall to 0.746 and 0.592 for $10^5 M_\odot$ seeds. At $\epsilon = 0.05719$, light seeds require 0.793 and 0.740. For Eddington-limited active phases separated by zero-accretion phases, these imply active fractions of about 79 and 74 per cent of the available time, requiring a sustained fuel supply.

GN-z11's reference requirement is 1.437. Seeding at $z = 20$ raises it to 1.880, above UNCOVER-20466's 1.733, reversing their order. A mass measured with a method comparable to the Balmer-line sample would test this timing constraint. GN-z11 meets the primary evidence criterion and is placed in the exploratory sample because it uses a UV mass estimator.

5.4 Limitations

The contributing studies use different selection criteria and detection limits. Inferring population demographics would require a combined completeness model. Virial calibration and geometry errors are incompletely reported. The common mass offsets test a uniform shift; biases specific to individual sources, correlated errors and selection-dependent scatter remain unmodelled.

Table 6: Mass-estimator limitations and proposed observations for all 12 primary objects above the reference threshold. Caveats summarize the source records cited in Table 1; proposed observations target the stated limitations; their feasibility has not been assessed.

Object	Mass-estimator caveat	Proposed observation
UNCOVER-20466	Hbeta; mass depends on lens model and BLR interpretation.	Independent lens model and broad-line profile constraints.
GS-20057765	Hbeta; extinction and calibration uncertainties are incompletely quantified.	Deeper line profiles and an extinction constraint.
COSMOS3D-13852	Hbeta; source warns of possible 1–2 dex scattering bias.	Separate scattering wings from virial broadening.
RUBIES-EGS-55604	Hbeta; dust-corrected mass; separate 0.5 dex calibration scale.	Test dust correction and broad-line kinematics.
CEERS-1019	Hbeta; broad component has comparatively low significance.	Confirm the broad component with deeper spectroscopy.
GS-20030333	Hbeta; large quoted errors and uncertain extinction and calibration.	Improve line profile and extinction constraints.
GS-164055	Hbeta; reported interval straddles the reference threshold.	Reduce mass uncertainty with deeper spectroscopy.
GN-38509	Halpha; separate 0.3 dex calibration scale.	Independent estimator or calibration constraint.
J0910_2028_12910	Halpha; 0.01 dex formal error excludes 0.5 dex mass scale.	Measure the mass with an independent calibration.
CEERS-00717	Halpha; local calibration; no numeric systematic supplied.	Cross-check the virial estimator and line decomposition.
ZS7	Hbeta; quoted 0.4 dex already includes calibration scatter.	Spatially separate nuclear kinematics and interacting components.
GN-4685	Hbeta; reported interval straddles the reference threshold.	Improve line profile and extinction constraints.

Two duplicate pairs have been reconciled using the source papers; the identities of three groups remain unresolved (Appendix A.1). Excluding all six affected records preserves the threshold results. Reinstating these records requires checking their spectra, masses, images or apertures against the source papers. Counts of unique objects and hosts remain provisional.

Equation 2 describes growth through an average accretion rate, a fixed merger multiplier and constant radiative efficiency. Feedback and individual accretion histories remain unspecified. The probabilities hold these assumptions fixed. Averaging over them would require justified parameter distributions and correlations; identifying seed origins would also require formation models and population information.

Independent source checks cover all admitted mass/error fields and central redshifts. Coordinate checks and missing fields are documented with the release. Validation of redshift errors, full posteriors and auxiliary observables is incomplete. The catalogue covers the declared sources through 3 September 2026; other JWST discoveries may fall outside this source list.

6 Conclusions

For the adopted masses and growth assumptions, we find:

- For $100\,M_\odot$ seeds formed at $z = 30$, $\epsilon = 0.1$ and no merger contribution, 12 of the 224 objects in the main sample require an average Eddington ratio above one. Six do so in at least 95% of Monte Carlo draws from their quoted mass-error distributions.

- For UNCOVER-20466, COSMOS3D-13852 and RUBIES-EGS-55604, at least 95% of draws still require an average Eddington ratio above one after reducing the masses by 0.5 dex. Their mass estimates require tests of lensing, scattering and calibration, respectively.
- The assumed start of growth changes which objects require average accretion above the Eddington limit and which requires the highest rate. At constant efficiency $\epsilon = 0.05719$, every central mass in the 234-object expanded sample is reachable from a $100\,M_\odot$ seed at $z = 30$ with an average Eddington ratio below one.
- For A2744-QSO1, replacing the adopted virial mass with the independent dynamical estimate raises the required average Eddington ratio from 0.929 to approximately one. Its mass uncertainty allows average rates both below and above the Eddington limit.

Excluding the six records with unresolved identities preserves the threshold counts. Counts of unique objects in the full catalogue remain provisional. The proposed observations test the mass estimates of the most demanding objects under the stated growth assumptions.

Appendix A Source-family inventory

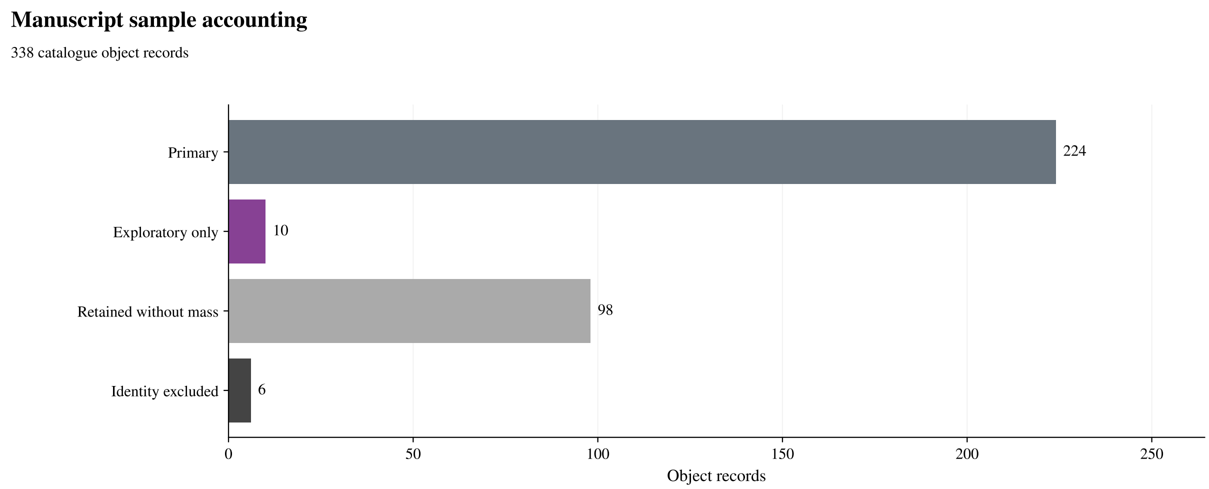

Figure 5: Sample accounting for 338 catalogue object records. Mutually exclusive categories comprise 224 primary objects, ten additional exploratory objects, 98 retained without eligible masses and six identity exclusions. The inclusive exploratory numerical sample contains 234 objects.

Table 7: Catalogue size and analysis samples.

Product or subset	Rows or objects
Measurements / provisional objects / hosts	350 / 338 / 337
Growth-eligible measurements / objects	244 / 237
Method-eligible primary measurements / objects	234 / 227
Manuscript primary / exploratory objects	224 / 234
Identity-excluded object records	6
Catalogue-only object records	101

Table 8 gives every admitted source family. We count measurements and preferred objects separately. Each preferred object is counted once under its adopted source. Auxiliary coordinates are attributed in the registry to THRILS (T. A. Hutchison et al. 2025), UHZ1 spectroscopy (A. D. Goulding et al. 2023), and the versioned JADES DR3 GOODS-S prism catalogue. The machine-readable companion retains full selection-channel and caveat tags alongside source URLs. “Context” and “Limit” denote values retained outside numerical growth inference.

Table 8: Source-family inventory. Counts are admitted measurements / eligible mass measurements / preferred primary objects after identity exclusion (M/G/P). The columns sum to 350/244/224; G includes alternatives, whereas P counts each selected object once. “Balmer” denotes single-epoch virial masses.

Source	M/G/P	Mass	Selection channel; principal caveat
C. J. Baccus & X. Xu (2026)	49/49/48	Balmer	NIRSpec broad lines; Frozen preprint masses; calibration
Ó. A. Chávez Ortiz et al. (2026)	1/0/0	Context	UV/optical diagnostics; Model-dependent mass
J. Chisholm et al. (2024)	1/0/0	None	High-ionization lines; No canonical mass

Table 8 continued

Source	M/G/P	Mass	Selection channel; principal caveat
K. Davis et al. (2026) / THRILS	6/6/6	Balmer	EELG-selected spectroscopy; Virial calibration
Q. Fei et al. (2026) / GLIMPSE	10/10/8	Balmer	Lensed broad lines; Lensing corrections
J. E. Greene et al. (2024) / UNCOVER	9/9/9	Balmer	Lensed broad lines; Lens model; BLR interpretation
Y. Harikane et al. (2023)	10/10/5	Balmer	NIRSpec broad lines; Local virial calibration
I. Juodžbalis et al. (2026a) / JADES	23/23/21	Balmer	Broad Balmer lines; Extinction; calibration
M. Killi et al. (2024)	1/1/1	Balmer	Broad Halpha; Linear mass-error conversion
D. D. Kocevski et al. (2025) / LRDs	6/6/6	Balmer	Broad Balmer lines; Dust correction; calibration
R. L. Larson et al. (2023) / CEERS-1019	1/1/1	Balmer	Broad Hbeta; Broad-component significance
X. Lin et al. (2024) / ASPIRE	16/16/16	Balmer	Broad Halpha; Virial calibration
X. Lin et al. (2026) / COSMOS-3D	13/13/13	Balmer	NIRCam grism broad lines; Possible scattering bias
J. Lyu et al. (2024) / SMILES	19/0/0	None	MIRI SED selection; SED-dependent AGN identification
R. Maiolino et al. (2024) / GN-z11	1/1/0	UV	UV lines; UV estimator outside primary
S. Mascia et al. (2026)	8/0/0	None	Compact blue broad emitters; No canonical mass
J. Matthee et al. (2024) / EIGER-FRESCO	20/20/19	Balmer	Broad Halpha; Dust; virial calibration
G. Mazzolari et al. (2025)	25/0/0	None	Narrow-line diagnostics; Star-formation overlap
G. C. K. Leung et al. (2026) / MEOW	12/0/0	None	MIRI SED selection; SED-dependent AGN fraction
R. P. Naidu et al. (2026) / MoM-BH*-1	1/0/0	Context	Broad Hbeta / SED; Scattering; stellar alternative
L. Napolitano et al. (2025a) / GHZ9	1/0/0	Context	UV lines / X-ray evidence; Mass assumes Eddington rate
L. Napolitano et al. (2025b) / GHZ4,7	2/0/0	None	Narrow / high-ionization lines; Tentative diagnostics
W. Ren et al. (2025) / ALPINE-CRISTAL	7/7/1	Balmer	Host-selected broad Halpha; Conditional BLR interpretations
J. Scholtz et al. (2025) / JADES	21/0/0	None	Narrow / high-ionization lines; Tentative classifications
D. D. Kocevski et al. (2026) / Skyfire (CEERS)	22/22/22	Balmer	Broad Halpha; Virial calibration
M. Tang et al. (2025)	3/0/0	None	High-ionization lines; Ionizing-source ambiguity
A. J. Taylor et al. (2025) / CEERS-RUBIES	37/37/35	Balmer	Broad Halpha; Slit loss; dust; calibration
H. Treiber et al. (2025) / UNCOVER	2/0/0	None	UV-line diagnostics; Stellar alternatives
H. Übler et al. (2024) / ZS7	1/1/1	Balmer	Offset broad Hbeta; Error includes calibration scatter
Á. Bogdán et al. (2024); F. Zou et al. (2026) / UHZ1	2/0/0	Context	X-rays / SED; Later detection disputed
Z. Zhang et al. (2026) / LRDs	5/0/0	Limit	Narrow lines / SED; Mass upper limits
M.-Y. Zhuang et al. (2026) / NEXUS	15/12/12	Balmer	NIRCam WFSS broad lines; Twelve masses lack errors

A.1 Preferred measurements and identity exclusions

The measurement–object link table records the source, publication version, quoted uncertainty and reason for each preferred measurement. The three unresolved identity groups are Baccus GDS_1210_9515 versus JADES GS-8083, GS-10013704 versus Scholtz 99671, and Scholtz 16745 versus 208643. All six records are excluded from manuscript inference. Three have eligible masses and otherwise pass the primary cuts; the other three lack eligible masses. Their spectra, images and apertures require reconciliation before readmission. Detailed admission and revision records accompany the catalogue release.

Appendix B Growth-model derivations and implementation

B.1 Accretion and exponential growth

The growth constraints follow from the relation between accretion luminosity, retained mass, and available cosmic time. For electron-scattering opacity in ionized hydrogen, the Eddington luminosity and luminosity-based Eddington ratio are

$$L_{\rm Edd} = \frac{4\pi G M_{\rm BH} m_p c}{\sigma_T}, \qquad f_{\rm Edd} = \frac{L_{\rm bol}}{L_{\rm Edd}}, \qquad t_{\rm Edd} = \frac{M_{\rm BH} c^2}{L_{\rm Edd}} = \frac{c\sigma_T}{4\pi G m_p} \simeq 0.45 \text{ Gyr}. \tag{4}$$

Here G, c, m_p, and σ_T denote the gravitational constant, speed of light, proton mass, and Thomson cross section. With radiative efficiency $0 < \epsilon < 1$ and rest-mass inflow rate $\dot{M}_{\rm in}$,

$$L_{\rm bol} = \epsilon \dot{M}_{\rm in} c^2, \qquad \dot{M}_{\rm BH} = (1-\epsilon)\dot{M}_{\rm in} = \frac{1-\epsilon}{\epsilon}\frac{f_{\rm Edd}}{t_{\rm Edd}} M_{\rm BH}. \tag{5}$$

The inflow rate is defined at the radiating flow; earlier mass loss is omitted. Dividing Eq. 5 by $M_{\rm BH}$ and integrating at fixed ϵ gives

$$\ln\left(\frac{M_{\rm BH}}{B_{\rm merge} M_{\rm seed}}\right) = \frac{1-\epsilon}{\epsilon}\frac{1}{t_{\rm Edd}} \int_{t(z_{\rm seed})}^{t(z_{\rm obs})} f_{\rm Edd}(t)\, dt = \frac{1-\epsilon}{\epsilon}\frac{\overline{f_{\rm Edd}}\,\Delta t}{t_{\rm Edd}}. \tag{6}$$

Exponentiating yields Eq. 2; solving for the average rate and applying the non-negative floor yields Eq. 3. For a constant positive $f_{\rm Edd}$, the e-folding time is

$$t_{\rm efold} = \frac{\epsilon}{1-\epsilon}\frac{t_{\rm Edd}}{f_{\rm Edd}}. \tag{7}$$

It is 0.05 Gyr at $\epsilon = 0.1$ and $f_{\rm Edd} = 1$.

B.2 Merger multiplier and inverse seed mass

The implementation evaluates Eq. 2 in log mass:

$$\log_{10}\left(\frac{M_{\rm BH}}{M_\odot}\right) = \log_{10}\left(\frac{M_{\rm seed}}{M_\odot}\right) + \log_{10} B_{\rm merge} + \frac{1-\epsilon}{\epsilon}\frac{\overline{f_{\rm Edd}}\,\Delta t}{t_{\rm Edd}\ln 10}. \tag{8}$$

We evaluate $B_{\rm merge} = 1$ (no boost) and 2 (a factor-of-two mass boost). At fixed accretion history,

$$\frac{M_{\rm BH}(B_{\rm merge}=2)}{M_{\rm BH}(B_{\rm merge}=1)} = 2, \qquad \Delta \log_{10} M_{\rm BH} = \log_{10} 2 \simeq 0.3010. \tag{9}$$

Solving the same relation for seed mass gives

$$\log_{10}\left(\frac{M_{\rm seed,req}}{M_\odot}\right) = \log_{10}\left(\frac{M_{\rm BH}}{M_\odot}\right) - \log_{10} B_{\rm merge} - \frac{1-\epsilon}{\epsilon}\frac{\overline{f_{\rm Edd}}\,\Delta t}{t_{\rm Edd}\ln 10}. \tag{10}$$

Here $M_{\rm seed,req}$ is the seed mass required for a specified average rate, efficiency, growth interval and merger multiplier.

B.3 Cosmic time

Cosmic ages use the flat matter-plus-Lambda relation

$$t(z) = \frac{2}{3H_0\sqrt{\Omega_\Lambda}}\,\mathrm{asinh}\left[\sqrt{\frac{\Omega_\Lambda}{\Omega_m}}(1+z)^{-3/2}\right], \tag{11}$$

with $H_0 = 67.3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315$, and $\Omega_\Lambda = 0.685$. The Hubble constant is converted to inverse time; ages and growth intervals are in Gyr. Radiation is neglected.

B.4 Two-state duty cycle

For a two-state history with the same efficiency in both states,

$$\overline{f_{\rm Edd}} = D f_{\rm burst} + (1-D) f_{\rm quiet}, \qquad D_{\rm req} = \frac{\overline{f_{\rm Edd}}_{\rm req} - f_{\rm quiet}}{f_{\rm burst} - f_{\rm quiet}}. \tag{12}$$

Here D is the active-state time fraction, $f_{\rm burst} > f_{\rm quiet} \geq 0$, and the diagnostic requires $\overline{f_{\rm Edd}}_{\rm req} \geq f_{\rm quiet}$. We retain values $D_{\rm req} > 1$ to flag histories that would require more active time than is available. The inferred duty cycle depends on the assumed burst and quiet-state rates.

B.5 Sampling, mass conversions and track selection

Draws use seed 20260808 and a stable identifier-based random stream. For 10,000 draws, the binomial Monte Carlo standard error of a probability is at most 0.005 (approximately 0.0022 at probability 0.95). This numerical error measures the precision from a finite number of draws. The twelve mass-bearing NEXUS objects have no published mass errors: repeated point values produce zero-width stored intervals, while probabilities and uncertainty ranks remain unavailable. Table 1 identifies separately reported calibration scales and scatter already included in the quoted errors.

The mass-scale tests add the same fixed offset to every original mass draw. The final published Killi mass $8^{+0.5}_{-0.4} \times 10^8 M_\odot$ is converted by transforming both linear bounds, giving $\log_{10}(M/M_\odot) = 8.90309^{+0.02633}_{-0.02228}$.

For Figure 1, candidate average rates span 0.1–2.0 in steps of 0.1. For each seed mass and fixed efficiency, we retain the two rates with the most primary objects within 0.5 dex of either the $B = 1$ or $B = 2$ prediction, requiring at least five such objects. Ties favour smaller median absolute residuals, then smaller rates. Most objects selected by this 0.5-dex plotting tolerance lie at $z \simeq 4$–6. Baseline rankings, duty-cycle diagnostics and seed-redshift–mass maps retain $\epsilon = 0.1$. Scientific conclusions use the required-rate ranking.

Appendix C Compatibility across seed masses and accretion rates

C.1 Efficiency prescription

For the illustrative spin-dependent maps, the ideal Kerr thin-disk efficiency is the ISCO binding efficiency (J. M. Bardeen et al. 1972):

$$\epsilon_a = 1 - \sqrt{1 - \frac{2}{3r_{\rm ISCO}}}, \tag{13}$$

$$r_{\rm ISCO} = 3 + Z_2 - \mathrm{sgn}(a)\sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}, \tag{14}$$

$$Z_1 = 1 + (1 - a^2)^{1/3}\left[(1 + a)^{1/3} + (1 - a)^{1/3}\right], \qquad Z_2 = \sqrt{3a^2 + Z_1^2}. \tag{15}$$

The dimensionless spin is $a = cJ/(GM_{\rm BH}^2)$, where J is the signed angular momentum relative to the disk, and $r_{\rm ISCO}$ is in units of $GM_{\rm BH}/c^2$. The ideal efficiencies at $a = -1, 0, +1$ are approximately 0.038, 0.057, and 0.423, respectively, neglecting photon capture and stress at the ISCO. Logarithmic luminosity growth at supercritical inflow is motivated by advection and wind models (J. Poutanen et al. 2007). We use a simplified logarithmic relation with coefficient one and omit radial mass loss. For Eddington ratios above unity, the maps adopt

$$f_{\rm Edd} = 1 + \ln \dot{m}, \qquad \dot{m} = \frac{\dot{M}_{\rm in}}{L_{\rm Edd}/(\epsilon_a c^2)}, \qquad \epsilon_{\rm eff} = \begin{cases} \epsilon_a, & f_{\rm Edd} \le 1, \\ \epsilon_a f_{\rm Edd} \exp(1 - f_{\rm Edd}), & f_{\rm Edd} > 1. \end{cases} \tag{16}$$

The logarithmic luminosity relation applies only for super-Eddington accretion. Each compatibility cell holds both the rate and its effective efficiency constant over the interval. The grid tests constant accretion states with fixed spin and omits relativistic slim-disc dynamics. For time-varying efficiency, growth depends on $\int [(1 - \epsilon(t))/\epsilon(t)] f_{\rm Edd}(t)\, dt$, which requires the joint time dependence of efficiency and accretion rate.

Figure 6 gives the full scenario grid. A primordial interpretation of the broad seed-mass interval requires additional assumptions about seed formation. The descriptive fractions use point masses and the inclusive interval criterion in Section C.2, without error integration. The Supplementary Material discusses this interval for pre-existing primordial black holes and illustrates an earlier start of accretion.

C.2 Seed-mass ranges and compatible fractions

For each assumed accretion rate, efficiency, starting redshift and merger contribution, we solve Eq. 10 for the initial seed mass that would grow to the observed black-hole mass. We then check whether that seed falls in each of four mass ranges: 10–100 $M_\odot$ (light), 10^3–10^4 $M_\odot$ (intermediate), 10^4–10^6 $M_\odot$ (heavy), and 10^2–10^6 $M_\odot$ (broad). We count an object as compatible with a range when some seed within it can reproduce the observed mass under these assumptions. The grid uses $z_{\rm seed} = 30$, three spin/efficiency cases, merger multipliers 1 and 2, and average rates 0.3, 1 and 2. These maps use the supercritical efficiency prescription in Section C.1.

The seed-mass ranges overlap and are tested separately, without probability weights. The broad range spans several possible seed-formation channels. If the seed mass needed to reproduce an object is smaller than the lower bound of a tested range, even the smallest seed in that range would grow larger than the observed black hole at the assumed accretion rate. That object is therefore excluded from the count for that range and set of growth assumptions. For each set of assumptions, we report the percentage of catalogue objects whose observed masses can be reproduced from a seed in the tested range. These percentages describe this literature sample. Estimating the corresponding percentages among all high-redshift black holes would require accounting for how each contributing study selected and detected its objects.

The primary and exploratory samples show similar compatibility patterns (Fig. 6). For heavy seeds at $a = +1$ and $f = 1$, changing B_{merge} from one to two raises compatibility from 57 to 70 per cent in the primary sample (59 to 71 per cent inclusive). At higher accretion rates, even the smallest seed in a tested mass range can grow larger than the observed black hole. Such objects reduce the percentage reproduced from seeds in that range. These fractions use central mass estimates and the efficiency prescription in Section C.1.

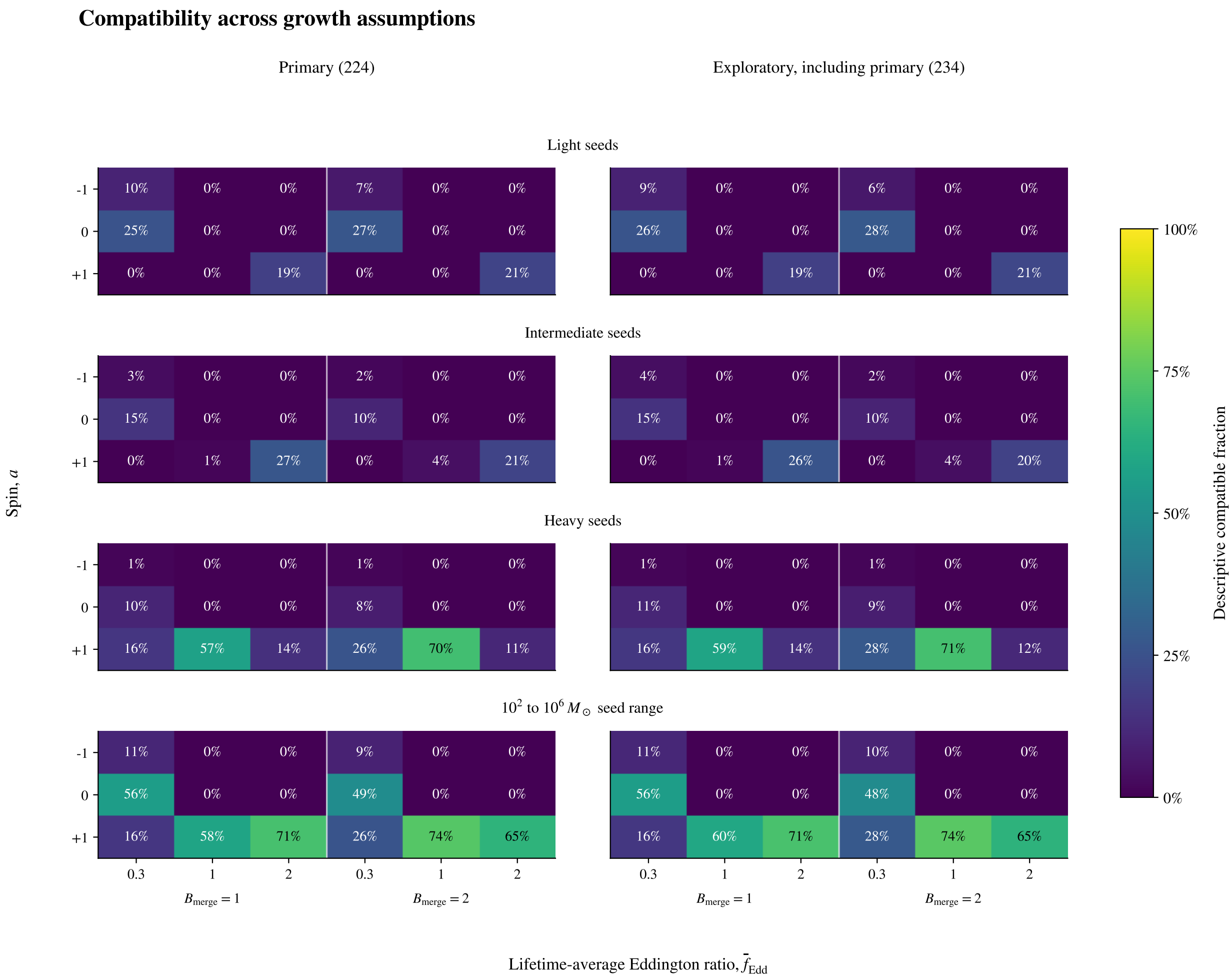

Figure 6: Constant-state compatibility fractions in the primary sample (left) and the inclusive exploratory sample (right). The exploratory sample includes all primary objects. Each column specifies merger multiplier B and fixed luminosity ratio f; rows specify ideal spin-dependent efficiency, with the supercritical prescription in Section C.1. Seed intervals overlap and have no assigned probability weights.

Appendix D Data and code availability

The catalogue, source registries, code, and manuscript analysis are available at https://github.com/nniickels/highz-accretion-atlas. The frozen v3 catalogue baseline is identified by Git commit a40a0d2; version manifests under `releases/` record the dataset artifact hashes. Comparison results and tables are in `paper/analysis/`. External inputs are in `paper/review_inputs.json`. An archival DOI is pending.

The numbered notebooks reproduce the catalogue, analysis, figures and atlas using pinned dependencies.

Instructions and verification procedures are in `docs/guides/reproducibility.md`. The source-family inventory in Appendix A links measurements, methods, selection channels, and major caveats. Full identifiers, source-native observables, alternative measurements, and per-object maps remain available.

These checks verify transcription and numerical reproduction. The validity of the mass estimators and unresolved object identities require observational assessment. All records in the three unresolved groups listed in Section 5.4 are excluded from manuscript inference.

References

Baccus, C. J., & Xu, X. 2026, ApJ, 1006, 165, doi: 10.3847/1538-4357/ae7de7

Bardeen, J. M., Press, W. H., & Teukolsky, S. A. 1972, ApJ, 178, 347, doi: 10.1086/151796

Bogdán, Á., Goulding, A. D., Natarajan, P., et al. 2024, Nature Astronomy, 8, 126, doi: 10.1038/s41550-023-02111-9

Chávez Ortiz, Ó. A., Finkelstein, S. L., Plat, A., et al. 2026, ApJ, 1008, 227, doi: 10.3847/1538-4357/ae96a4

Chisholm, J., Berg, D. A., Endsley, R., et al. 2024, MNRAS, 534, 2633, doi: 10.1093/mnras/stae2199

Davis, K., Brooks, M., Trump, J. R., et al. 2026, ApJ, submitted. https://arxiv.org/abs/2602.23310v1

Dayal, P. 2024, A&A, 690, A182, doi: 10.1051/0004-6361/202451481

Fei, Q., Fujimoto, S., Naidu, R. P., et al. 2026, ApJ, 1003, 244, doi: 10.3847/1538-4357/ae6248

Goulding, A. D., Greene, J. E., Setton, D. J., et al. 2023, ApJL, 955, L24, doi: 10.3847/2041-8213/acf7c5

Greene, J. E., Labbe, I., Goulding, A. D., et al. 2024, ApJ, 964, 39, doi: 10.3847/1538-4357/ad1e5f

Harikane, Y., Zhang, Y., Nakajima, K., et al. 2023, ApJ, 959, 39, doi: 10.3847/1538-4357/ad029e

Hutchison, T. A., Larson, R. L., Arrabal Haro, P., et al. 2025, arXiv e-prints. https://arxiv.org/abs/2512.12509v1

Juodžbalis, I., Maiolino, R., Baker, W. M., et al. 2026a, MNRAS, 546, stag086, doi: 10.1093/mnras/stag086

Juodžbalis, I., Marconcini, C., D'Eugenio, F., et al. 2026b, Nature, 653, 1017, doi: 10.1038/s41586-026-10579-4

Killi, M., Watson, D., Brammer, G., et al. 2024, A&A, 691, A52, doi: 10.1051/0004-6361/202348857

Kocevski, D. D., Finkelstein, S. L., Barro, G., et al. 2025, ApJ, 986, 126, doi: 10.3847/1538-4357/adbc7d

Kocevski, D. D., Taylor, A. J., Onoue, M., et al. 2026, ApJ, submitted. https://arxiv.org/abs/2609.00112v1

Larson, R. L., Finkelstein, S. L., Kocevski, D. D., et al. 2023, ApJL, 953, L29, doi: 10.3847/2041-8213/ace619

Leung, G. C. K., Eilers, A.-C., Endsley, R., et al. 2026, ApJ, submitted. https://arxiv.org/abs/2607.02666v1

Lin, X., Fan, X., Wang, F., et al. 2026, ApJ, 996, 93, doi: 10.3847/1538-4357/ae1b9b

Lin, X., Wang, F., Fan, X., et al. 2024, ApJ, 974, 147, doi: 10.3847/1538-4357/ad6565

Lyu, J., Alberts, S., Rieke, G. H., et al. 2024, ApJ, 966, 229, doi: 10.3847/1538-4357/ad3643

Maiolino, R., Scholtz, J., Witstok, J., et al. 2024, Nature, 627, 59, doi: 10.1038/s41586-024-07052-5

Mascia, S., Matthee, J., Torralba, A., et al. 2026, A&A, submitted. https://arxiv.org/abs/2608.25021v1

Matthee, J., Naidu, R. P., Brammer, G., et al. 2024, ApJ, 963, 129, doi: 10.3847/1538-4357/ad2345

Mazzolari, G., Scholtz, J., Maiolino, R., et al. 2025, A&A, 700, A12, doi: 10.1051/0004-6361/202451860

Naidu, R. P., Matthee, J., Katz, H., et al. 2026, Nature, 656, 329, doi: 10.1038/s41586-026-10846-4

Napolitano, L., Castellano, M., Pentericci, L., et al. 2025a, ApJ, 989, 75, doi: 10.3847/1538-4357/ade706

Napolitano, L., Castellano, M., Pentericci, L., et al. 2025b, A&A, 693, A50, doi: 10.1051/0004-6361/202452090

Poutanen, J., Lipunova, G., Fabrika, S., Butkevich, A. G., & Abolmasov, P. 2007, MNRAS, 377, 1187, doi: 10.1111/j.1365-2966.2007.11668.x

Ren, W., Silverman, J. D., Faisst, A. L., et al. 2025, MNRAS, 544, 211, doi: 10.1093/mnras/staf1709

Scholtz, J., Maiolino, R., D'Eugenio, F., et al. 2025, A&A, 697, A175, doi: 10.1051/0004-6361/202348804

Shen, Y. 2013, Bulletin of the Astronomical Society of India, 41, 61. https://arxiv.org/abs/1302.2643v1

Tang, M., Stark, D. P., Plat, A., et al. 2025, ApJ, 991, 217, doi: 10.3847/1538-4357/adfd57

Taylor, A. J., Finkelstein, S. L., Kocevski, D. D., et al. 2025, ApJ, 986, 165, doi: 10.3847/1538-4357/add15b

Treiber, H., Greene, J. E., Weaver, J. R., et al. 2025, ApJ, 984, 93, doi: 10.3847/1538-4357/adc38f

Übler, H., Maiolino, R., Pérez-González, P. G., et al. 2024, MNRAS, 531, 355, doi: 10.1093/mnras/stae943

Zhang, Z., Jiang, L., Liu, W., Ho, L. C., & Inayoshi, K. 2026, ApJ, 998, 170, doi: 10.3847/1538-4357/ae33c4

Zhuang, M.-Y., Li, J., Shen, Y., et al. 2026, ApJ, 999, 31, doi: 10.3847/1538-4357/ae3612

Zou, F., Gallo, E., Zuo, Z., et al. 2026, ApJ, in press. https://arxiv.org/abs/2603.24893v2